

Output Form (OF)

Score: 3/5 — Moderate gaps | Confidence: medium

Benchmark: DRBENCHMARK | Context: French Pharmaceutical Regulatory Affairs — Document Compliance NLP

Definition: Whether the expected output modality matches regional deployment needs and accessibility.

Justification

The benchmark's output forms — discrete labels (NER tags, classification labels) and continuous similarity scores [Q22, Q41, Q52, Q54] — are broadly compatible with the deployment's multi-candidate confidence-score and human-in-the-loop adjudication design. Models are evaluated with consistent protocols (4-run averaging, Student's t-test [Q51], SeqEval IOB2 [Q52], Spearman/EDRM for STS [Q54]). However, the benchmark does NOT evaluate or document confidence calibration, uncertainty quantification, or score thresholds for triggering mandatory human review — operationally critical for the deployment's borderline-case escalation workflow. Leading model performance varies by category, with no model excelling on both STS and MLC simultaneously [Q82], indicating that single-model deployment for compliance checking would have inconsistent output reliability. The 0–5 STS scale is continuous and structurally adequate but provides no calibration data for legal-equivalence threshold setting.

Strengths

- Discrete label and continuous score outputs are structurally compatible with multi-candidate confidence-score deployment design [Q22, Q52]
- Statistical significance reporting via 4-run averaging and t-test [Q51]
- Tokenizer-agnostic SeqEval IOB2 evaluation provides consistent NER scoring [Q52, Q53]
- Inter-annotator score arrays in DEFT2020 (DEFT2020-D7) provide raw annotator variance data, in principle usable for human-review threshold calibration
- Hyperparameter and reproducibility documentation [Q49, Q99]

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality (text labels and continuous scores) matches deployment needs (NER spans + STS scores in document compliance system). No modality mismatch.

OF-2: Text-to-speech not applicable — deployment is text-only document processing for regulatory specialists.

OF-3: Literacy/accessibility not applicable in the standard sense — the user population consists of regulatory affairs specialists, pharmacologists, legal experts (high domain expertise).

OF-4: Form mismatches relative to deployment: (a) confidence calibration not benchmarked — no probability calibration metrics, expected calibration error, or threshold semantics provided; (b) score-threshold for human-review escalation not documented in any dataset; (c) no model excels uniformly across STS and MLC [Q82], indicating output reliability varies by task category — relevant for compliance tasks that span multiple sub-tasks; (d) tabular/abbreviated data outputs (CLISTER-D19, CLISTER-D20) atypical of regulatory prose. These are external validity concerns: model outputs in deployment

may be miscalibrated for the borderline-case adjudication that the operational workflow depends on.

ID	Evidence
DEFT2020-D7	scores: [5.0, 2.0, 4.0, 4.0, 5.0] — Wide annotator spread on safety instruction; relevant to human adjudication design
CLISTER-D19	Jour 45 - - - - - 25 µg 0,4 mg" / "Jour 53 15,13 12,0 1,58 - - - - - 25 µg 0,4 mg — Table rows with dash placeholders; structurally unlike regulatory prose
PXCORPUS-D18	non prendre 150 milligrammes par jour — One of three replace examples sampled; class severely underrepresented

Information Gaps

- Whether deployment system documentation specifies STS score thresholds for mandatory human review
- Whether deployment includes confidence calibration layer external to the underlying model

Requires Expert Verification

- Operational thresholds for borderline-case escalation in regulatory workflows
- Whether deployment monitors task-category-specific performance (STS vs. NER vs. MLC) given that no single model excels uniformly

Remediation

Gap 1: Confidence calibration, uncertainty quantification, and human-review score thresholds not benchmarked, despite deployment's human-in-the-loop borderline-case adjudication design

Recommendation: Add calibration evaluation (expected calibration error, reliability diagrams) on STS and NER tasks. Document score thresholds at which human review should be triggered, calibrated against regulatory-affairs annotator disagreement rates (using DEFT2020 inter-annotator score arrays as a starting point for variance analysis).